MILITARY SPECIFICATION

REGULATOR, CURRENT, RUNWAY LIGHTING, 120 VOLT, 2-1/4 KW, AC

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 120 volt, 2-1/4 kW, constant-current regulator for use with airfield runway lighting circuits.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-R-14328	- Rubber, Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

FSC 6110

beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions For Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-810	- Environmental Test Methods.
MS27289	- Transformer, Power, Isolation-Airport Lighting, 30-45W/6.6A/6.6A, 5000V.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

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3. REQUIREMENTS

3.1 Description. The 120 volt, 2-1/4 kW runway lighting current regulator (hereinafter called "regulator") shall consist of a control switch, open circuit protector, and a resonant circuit constant-current network encased in a steel tank or cubicle.

3.2 First article (preproduction model). The contractor shall furnish one regulator for examination and testing within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce regulators that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion caused by fuels, salt spray, or atmospheric conditions as specified in 3.6.

3.3.1.1 Protective treatment. When materials are used in the construction of the regulator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.2 Use of commercial parts. Bolts, nuts, washers, cotter pins, lock rings and similar fastening devices used for assembly, may be selected from commercial sources, provided military or other standard parts are not specifically called out by this specification or associated standards. If such commercial parts are used, they shall possess suitable properties and shall be replaceable by military standard parts without alteration, and provided the corresponding military standard part numbers are referenced in the parts list, and, if practicable, on the contractor's drawings.

3.4 Design. The regulator shall be designed to supply a 2-1/4 kW, ac, 6.6-ampere series lighting load, characteristic of portable series-type runway lighting circuits, from a 100 to 120 volt, two-wire, single-phase, 60-Hertz, constant-voltage source. The regulator shall be capable of supplying the connected load at 6.6, 5.5, 4.8, 4.2, and 3.8 amperes.

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3.4.1 Type. The regulator shall be of the self-cooled, dry type, but component parts may be mineral oil insulated, provided that each part so insulated is permanently sealed. The entire regulator shall be sealed to permit outdoor use under the environmental conditions specified in 3.4.13.

3.4.2 Fuse. A standard commercial 250 volt, 3-ampere fuse and fuse block shall be located inside the tank. The fuse shall be connected between the X1 bushing and the control switch.

3.4.3 Open-circuit protector. An open-circuit protective device shall be supplied that will open the X1 line within 2 seconds in the event that the load circuit is opened. The device shall be reset by turning the control switch off, then on. The contactor which opens the X1 line shall have a current-carrying capacity of not less than 30 amperes.

3.4.4 Input taps. Five input taps for nominal voltages of 100, 105, 110, 115, and 120 V shall be provided. The taps shall be connected to an internal terminal board or tap changer in such a manner that the taps can easily be changed after the tank cover is removed. Unless otherwise specified, the regulator shall be connected for 110 V (nominal), when shipped. A tolerance of ± 1.5 V will be allowed on the 100, 105, 110, and 115 V taps, provided that they are labeled with the actual design voltage, rather than the nominal voltage.

3.4.5 Control switch. A six-position rotary selector switch labeled "Off, Brightness 1, 2, 3, 4, 5" shall be provided. The control switch handle shall be located on the front or top of the regulator tank, and shall be recessed or shielded in order to prevent damage to the handle.

3.4.6 Brightness taps. The five brightness taps shall be connected directly to the control switch. The input and brightness taps may be placed on either the same transformer coil or on different coils.

3.4.7 Resonant-circuit network. The constant-current network shall have no moving parts and shall utilize a resonant circuit to produce a constant current of the values specified herein for the output when the proper voltages are supplied. The open-circuit voltage shall not exceed 200 percent of the rated load voltage. The design shall be such that the input circuit is completely isolated from the output circuit.

3.4.8 Circuit design. Any suitable control and constant-current circuit may be utilized, provided it meets the requirements of 3.4.7 and 3.6. Tubes or other parts requiring periodic replacement shall not be used in the regulator.

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3.4.9 Tank. All parts shall be housed in a sealed, rust-resistant, sheet steel, self-supporting container, except that the edges of the laminated core of the transformer may be exposed for cooling purposes provided the exposed edges are protected to prevent corrosion. The exact shape is optional, provided the switch handles are at a convenient operating height, and the unit is sufficiently stable to eliminate the need for bracing or fastening to the floor.

3.4.10 Terminals. The tank shall be provided with three porcelain stud terminals and one ground terminal. The one porcelain terminal and the ground terminal utilized on the input shall be labeled "Input, X1 and XN," respectively, and the two output porcelain terminals shall be labeled "Output, H1 and H2." The terminals shall be protected from damage by recessing or shielding. Two lifting lugs shall be welded to the outside of the tank. "Danger-High Voltage" shall be painted or stenciled on or near the output terminals.

3.4.11 Cover. Access to relays and input tap changes shall be obtained by means of a gasketed removable cover. The cover gasket shall be rubber conforming to MIL-R-14328. Since the regulator may be transported at high altitude, the tank shall be designed to withstand an internal pressure of 5 psi, and the following shall be painted or stenciled on the tank:

CAUTION: Loosen regulator cover before transporting at altitudes over 11,000 feet and reseal cover immediately after landing.

3.4.12 Wiring. The regulator shall be wired so that only input and output leads need be attached to permit operation.

3.4.13 Weather. The regulator shall be capable of continuous outdoor operation in temperatures from -65° F (-54° C) to +125° F (52° C) and shall not be damaged (see 6.4.1), by storage in temperatures from -65° F (-54° C) to +160° F (71° C).

3.4.14 Load. The regulator shall be capable of delivering 2.25 kW at 6.6 amp continuously.

3.4.15 Weight. The weight of the regulator shall not exceed 220 pounds.

3.5 Ease of maintenance. The regulator shall be constructed so that adjustment and repairs can easily be made by the personnel of operating units and overhaul bases without the use of special tools.

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3.6 Performance.

3.6.1 Environmental conditions.

3.6.1.1 Operational conditions. When tested as specified in 4.5.2.2, the regulator shall be capable of operating as specified in 3.6.4, without damage (see 6.4), under the following conditions:

(a) Temperature from -65° F to +125° F.
(b) Relative humidity up to 100 percent.
(c) Shock.

3.6.1.2 Storage and transit conditions. When tested as specified in 4.5.2.2, the regulator shall not be damaged by storage in temperatures from -65° F to 160° F.

3.6.2 Temperature rise. When tested as specified in 4.5.2.3, the temperature rise in continuous operation for any load from short circuit to full load shall not exceed 55° C for Class A insulation, 80° C for Class B insulation, and 140° C for Class H insulation.

3.6.3 Load. When connected to any of the specified supply voltages (design values), and with proper input voltage taps selected, the regulator shall meet the requirements of 3.6.3.1 - 3.6.3.3 when operated with a unity power factor load.

3.6.3.1 Brightness taps. If all taps are placed on the same coil, the arrangement shall be such as to produce the correct nominal-current values on all brightness taps when the 120V input tap is used, and the correct nominal-current values or brightness tap 5 when any one of the input taps is used.

3.6.3.2 Output current. At any load from short circuit to rated load, the output currents shall be within the limits specified in Table I.

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Table I. Output current limits

Tap	Nominal current (amps)	Full load resistance ohms	Output current tolerance (±) (in amps)		
			SC to 1/4 load	1/4 to 3/4 load	3/4 to full load
1	3.8	33	0.4	0.3	0.2
2	4.2	36	.4	.3	.2
3	4.8	41	.4	.3	.2
4	5.5	47	.4	.3	.2
5	6.6	51.7	.3	.2	.15

3.6.3.3 Overall efficiency and primary factor. The efficiency and power factor shall be not less than the values specified in Table II, when the regulator is operated at 120 V, Tap 5.

Table II. Efficiency and power factor values

Percent of rated kW output	Overall efficiency percent (minimum)	Primary power factor percent (minimum)
24	80	85
50	85	90
75	90	90
100	90	95

3.6.4 Isolating transformer load. When the regulator is connected to a 120 V, 60-cycle source and operated at room temperature, approximately 20° C, at loads consisting of lamps connected to the output circuit through isolating transformers which meet the electrical requirements of MS27289, the current shall not change more than the values specified in Table III, when the secondaries of the isolating transformers are opened.

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Table III. Current change due to opened transformer secondaries

Tap	Percent load	Maximum current change (±) in amps, when transformers are "open circuited"		
		Open-circuited transformers, percent		
		1-10 percent	11-20 percent	21-30 percent
1	50-100	0.4	0.7	1.0
2	50-100	.4	.7	1.0
3	50-100	.4	.7	1.0
4	50-100	.3	.5	.8
5	50-100	.2	.4	.6

3.6.5 Dielectric. When tested as specified in 4.5.2.8, the regulator shall withstand the application of the specified voltages without arcing, flashover, breakdown of insulation, or damage.

3.6.6 Control and protective device operation. When tested as specified in 4.5.2.9, all controls and protective devices shall operate satisfactorily without failure or damage within the time specified (see 3.4.3).

3.6.7 Leakage. When tested as specified in 4.5.2.10, there shall be no leakage or damage to the tank when specified pressures are applied.

3.7 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the regulator shall conform to MIL-STD-461, Class ID.

3.8 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnigh conforming to MIL-V-173, Composition I or II, as applicable (see 3.8.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

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(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.8.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.9 Slinging provisions. Permanently affixed slinging provisions that enable the regulator to be lifted in its normal operating position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 Nameplate. A nameplate, permanently and legibly marked, shall be attached to the regulator. The information marked in the spaces provided shall be in accordance with MIL-STD-130, and shall include:

REGULATOR, CURRENT
RUNWAY LIGHTING
2-1/4 KW
TYPE C-2
100/105/110/115/120 volts
6.6/5.5/4.8/4.2/3.8 amp.
60 cycle
Specification MIL-R-52939

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3.10.2 Instruction plate. A wiring diagram and instruction plate shall be attached to the outside of the regulator near the control switches. The instruction plate shall contain the following information in which slight modifications may be made or additional information furnished to provide complete instructions for the regulator being supplied:

CAUTION
DO NOT OPERATE REGULATOR WITH
OUTPUT CIRCUIT OPEN

Installation

1. Install regulator in as level a position as possible.
2. Check input bushings. Replace if cracked or broken.
3. Connect load circuit to output terminals.
4. Turn "Brightness Control" switch to "Off" position.
5. Check voltage of supply line at regulator, and set tap changer on regulator to match supply voltage.
6. Connect the supply leads to terminals X1 and X2, and the regulator is ready for operation.

Operation

1. Set the "Brightness Control" switch to the "On" position. The brightness of the lights should be set in accordance with the following:

Brightness setting for night operation

Brightness tap setting	Current value	Brightness	Visibility distance
		Percent	
1	3.80	1	Over 10 miles.
2	4.20	3	5 to 10 miles.
3	4.80	10	2 to 5 miles.
4	5.50	30	1 to 2 miles.
5	6.60	100	Less than 1 mile.

(The above values are approximate. Brightness of the lights should, when possible, be adjusted in accordance with radio request of pilot making landing.)

During daylight when visibility is 2 miles or less, the lights should be operated at 100 percent brightness. When visibility is greater than 2 miles, the lights are not needed.

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2. If the lights fail to operate or go out when the regulator is turned on, turn the switch to the "Off" position. Disconnect the load circuit and connect an ammeter directly across the output. (If no ammeter is available, a 6.6-amp lamp may be connected directly across the output to determine if the regulator is operating.) Turn the "Brightness Control" switch to "On." If The regular operates and the ammeter reads approximately 6.6 amp (or the lamp substituted for the ammeter burns at full brightness) with the brightness switch in the No. 5 position, the regulator is operating satisfactorily and the malfunctioning is due to an open or improper connection in the load circuit. If the ammeter reads low or high, or fails to read at all, (if a lamp is used and it appears very dim, very bright, or fails to light at all), turn the control switch to the "Off" position and make the following checks:

(a) Measure the voltage across the supply line and check the reading against setting of the input tap changer.
(b) Check connections to all terminals to determine:

(1) If leads are connected to correct terminals.
(2) If positive contacts are made in all connections.

(c) Check input bushings for cracked or broken porcelain.

3.11 Treatment and painting. All exterior (exposed) surfaces of the regulator shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color lusterless forest green. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.12 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts a defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.13 Workmanship.

3.13.1 General. The regulator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

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3.13.2 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw, bolt, or threads. All screw threads shall be in accordance with Handbook H-28.

3.13.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.13.4 Cleaning. The regulator shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction regulator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction regulator shall be tested as specified in 4.5.2.1 through 4.5.2.11. Failure of any test shall be cause for rejection.

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4.4 Quality conformance.

4.4.1 Examination. Each regulator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each regulator shall be tested as specified in 4.5.2.8 through 4.5.2.10. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The regulator shall be examined as specified herei for the defects listed in Table IV.

Table IV. Examination Schedule

Number	Characteristic	Requireme Paragrap
Major		
101	Dimensions that affect form, fit, or function.	3.1
102	Material not as specified.	3.3
103	Metals not as specified.	3.3.1
104	Protective treatment not as specified.	3.3.1.1
105	Use of commercial parts not as specified.	3.3.2
106	Type of regulator not as specified.	3.4.1
107	Fuse not as specified.	3.4.2
108	Open circuit protector not as specified.	3.4.3
109	Input taps not as specified.	3.4.4
110	Control switch not as specified.	3.4.5
111	Brightness taps not as specified.	3.4.6
112	Resonant-circuit network not as specified.	3.4.7
113	Circuit design not as specified.	3.4.8
114	Tank not as specified.	3.4.9

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Table IV. Examination Schedule (Cont'd)

Number	Characteristic	Requirement Paragraph
115	Terminals not as specified.	3.4.10
116	Cover not as specified.	3.4.11
117	Wiring not as specified.	3.4.12
118	Weight not as specified.	3.4.15
119	Fungus and moisture resistance not as specified.	3.8
120	Identification of product not in accordance with MIL-STD-130.	3.10
121	Instruction plate not as specified.	3.10.2
122	Interchangeability not as specified.	3.12
123	Screw assemblies and bolts not tight as specified.	3.13.2
124	Riveting not as specified.	3.13.3
Minor		
201	Dimensions that do not affect form, fit, or function.	3.1
202	Slinging provisions not as specified.	3.9
203	Nameplate not as specified.	3.10.1
204	Treatment and painting not as specified.	3.11
205	Workmanship on both the general work and the cleaning work not as specified.	3.13.1, 3.13.4

4.5.2 Tests.

4.5.2.1 Test conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, the test is to be made at atmospheric pressure (approximately 30 inches Hg) and at room temperature

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(approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.5.2.2 Environmental. The regulator shall be subjected to the followin tests of MIL-STD-810:

High Temperature	Method 501.1, Procedure II
Low Temperature	Method 502.1, Procedure I
Humidity	Method 507.1, Procedure I
Shock	Method 516.2, Procedure II

Nonconformance to 3.6.1 shall constitute failure of these tests.

4.5.2.3 Temperature rise. The temperature rise of the windings shall be determined by the resistance method. Nonconformance to 3.6.2 shall constitute failure of this test.

4.5.2.4 Regulation. The output current, when inspected at all five intensity steps with 30- and 100-percent load, shall be in accordance with Table I. This test shall be made at all input voltage ratings. Nonconformance to Table I shall constitute failure of this test.

4.5.2.5 Output current. The regulator shall be operated to determine that the output current is within the limits specified in 3.6.3.2. Nonconformance to 3.6.3.2 shall constitute failure of this test.

4.5.2.6 Overall efficiency and primary power factor. The regulator shal be operated at 120 volts, tap 5, to determine that the efficiencies and power factors are in accordance with 3.6.3.3. Nonconformance to 3.6.3.3 shall constitute failure of this test.

4.5.2.7 Isolating transformer load. The regulator shall be tested to determine compliance with 3.6.4. Nonconformance to 3.6.4 shall constitute failure of this test.

4.5.2.8 Dielectric. The circuits shall withstand the following 60-Hz root-mean-square voltages for 1 minute without failure:

(a) Input and control circuit to ground, 1,500 volts.
(b) Output circuit to primary and ground, 2,200 volts.

Nonconformance to 3.6.5 shall constitute failure of this test.

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4.5.2.9 Control and protective devices. All control and protective devices shall be tested for proper operation in accordance with 3.6.6. Nonconformance to 3.6.6 shall constitute failure of this test.

4.5.2.10 Leakage. The tank shall be tested for leakage by applying an internal pressure of 5 psi for 5 minutes. The air inlet shall then be sealed after the internal pressure has been reduced to atmospheric pressure. Nonconformance to 3.6.7 shall constitute failure of this test.

4.5.2.11 Electromagnetic interference. The regulator shall be tested to determine conformance to 3.7. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking of the regulators shall be examined and tested in accordance with the quality assurance provisions of MIL-E-17555.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. The regulator shall be preserved in accordance with the Level A requirements of MIL-E-17555.

5.1.2 Commercial. The regulator shall be preserved in accordance with the Level C requirements of MIL-E-17555.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. The regulator shall be packed in accordance with the Level A requirements of of MIL-E-17555.

5.2.2 Level B. The regulator shall be packed in accordance with the Level B requirements of MIL-E-17555.

5.2.3 Commercial. The regulator shall be packed in accordance with the Level C requirements of MIL-E-17555.

5.3 Marking. Marking shall be in accordance with MIL-E-17555 except marking for commercial packaging shall be in accordance with FED. STD. No. 356.

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6. NOTES

6.1 Intended use. The regulator is intended to be used for supplying power and current regulation to portable series-type airfield runway lighting circuits.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Degree of preservation and packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production regulators from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish regulators conforming to this specification.

6.4 Definitions. The following definition shall apply throughout this specification.

6.4.1 Damage. Damage is deformation, loosening, breakage, change of fit of any component or part, corrosion, or any condition resulting in malfunction of the regulator or any component or part.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each regulator.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
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Project 6110-A210

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-52939(ME) Regulator, Current, Runway Lighting, 120 Volt, 2-1/4 KW, AC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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